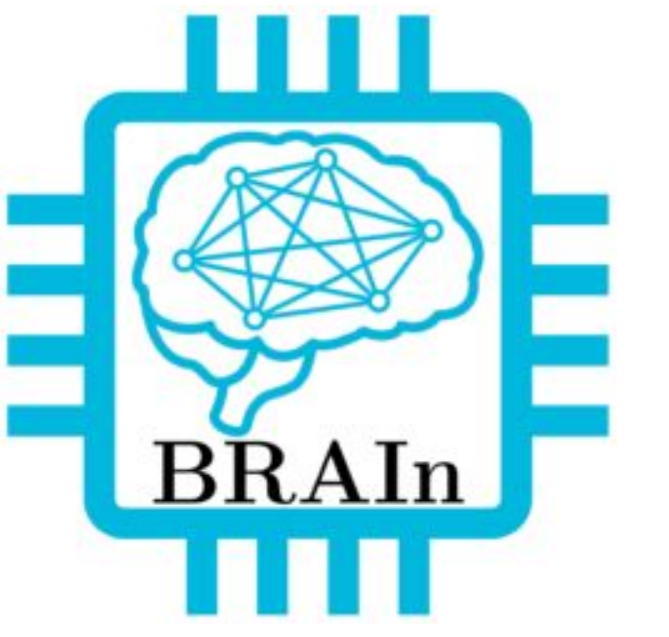




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Cross-Dataset Few-Shot Learning for Motor Imagery BCI Classification

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Introduction

Motivation

- ❖ BCI models are promising for many applications.
- ❖ Limitations that hinder their practical use:
 - Do not generalize well to new recording settings.
 - Require a calibration phase to adapt to a new user.

Related Work

1 Cross-Dataset

- ❖ Develop algorithms that can transfer knowledge between different subjects and EEG datasets with various recording setups.
- ❖ NeurIPS 2021 BEETL Challenge [1].

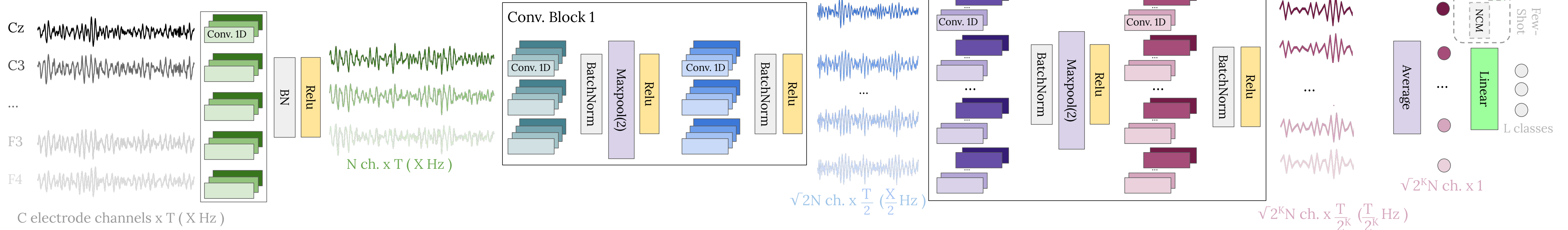
2 Few-Shot

- ❖ Sub-field of Machine Learning that consists of exploiting information from large datasets.
- ❖ In order to predict in domains or tasks where we have access to very little annotated data.

A simple and novel EEG Neural Network architecture : EEG-SimpleConv

Method

1 Architecture



2 Modern training ingredients

Normalization
Euclidean Alignment [2]

Training
Mixup + Subject-wise regularization

Testing
Session/Subject adaptation using BN

Within-Dataset Results

Performance per evaluation paradigm (% accuracy)

	Cross-Subject	Within-Subject	C.-S.+Fine-Tuning
(-EOG)	71.0	77.3	84.41
(+EOG)	79.6	81.9	90.19

Comparison to State-Of-The-Art Neural Networks

	Inception	TCNet	TIDNet	EEGNet*	ITNet	Our
(%)	63.5	65.1	65.4	66.2	69.4	71.0±0.2
(ms)	-	-	71.8	7.5	12.2	2.3

evaluated in the Cross-Subject paradigm
*reproduced results

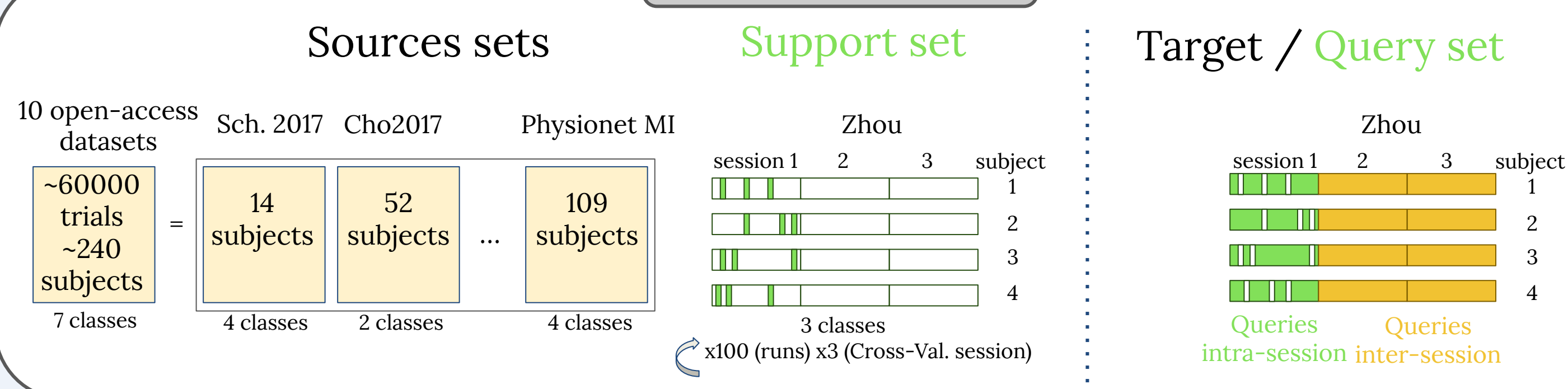
Performance per Dataset

	Cho2017	Zhou2016	PhysionetMI	BCI-IV2a
(%)	75.4	81.9	64.6	71.0
(n. classes)	2	3	4	4

evaluated in the Cross-Subject paradigm

Cross-Dataset Few-Shot

Method

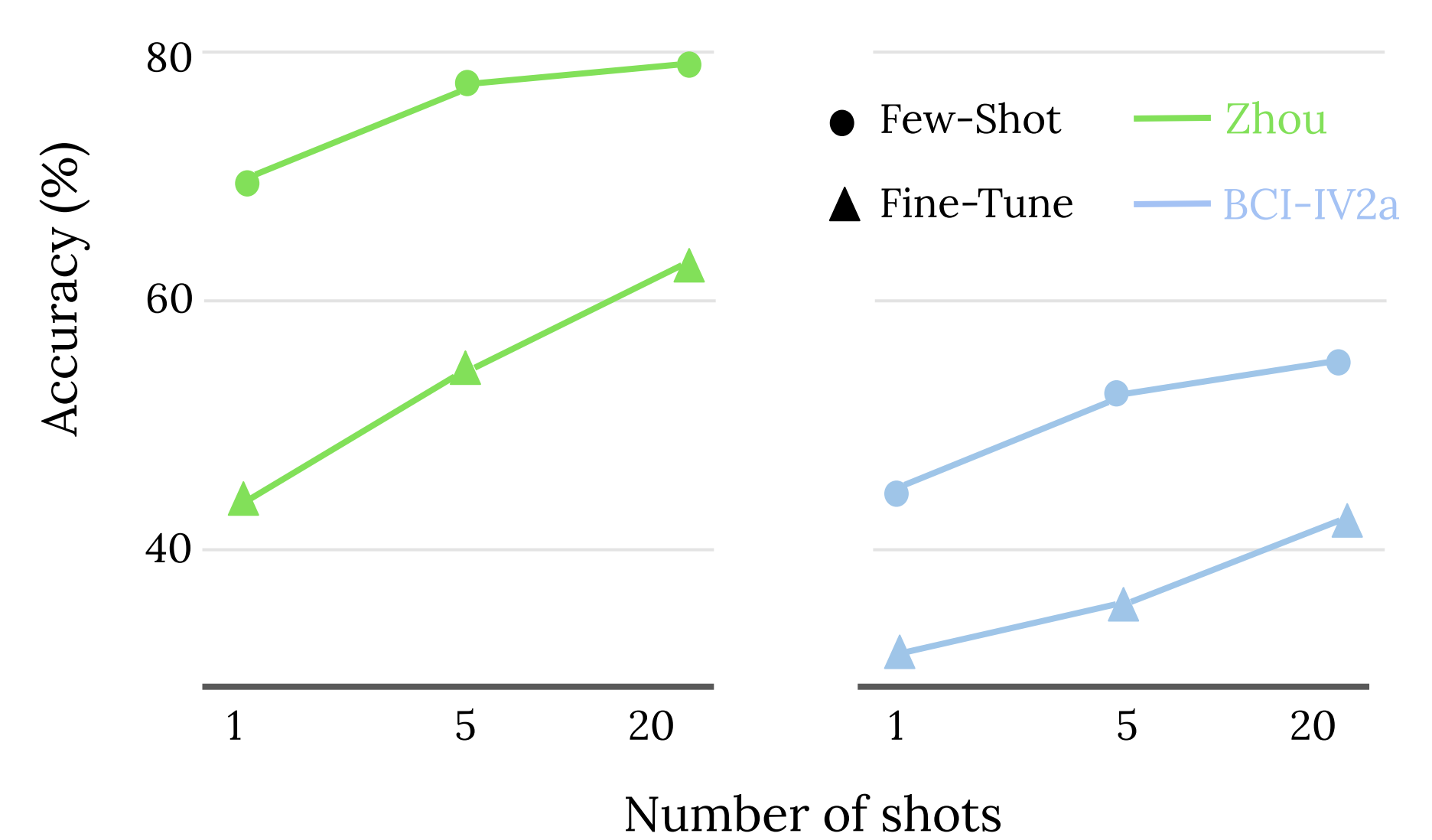


Detailed results

	BCI-IV2a			Zhou		
shots	1	5	20	1	5	20
Few-Shot						
(%) intra-session	44.6±0.6	52.3±0.3	55.3±0.2	70.0±0.8	77.6±0.5	79.6±0.4
(%) inter-session	44.2±0.6	51.6±0.3	54.6±0.2	68.4±0.7	76.5±0.4	78.2±0.2
Fine-Tune						
(%) intra-session	31.4±0.5	35.3±0.4	42.5±0.3	44.1±0.6	55.9±0.5	65.9±0.4
(%) inter-session	30.9±0.6	34.8±0.3	40.9±0.3	42.7±0.7	52.5±0.5	59.6±0.4

Results

Few Shot performances on Tests sets



- Leverage knowledge from multiple datasets is possible and useful.
- Few-Shot Learning methods can significantly improve performance.

Conclusion

- ❖ EEG-SimpleConv is a simple and efficient Neural Network architecture.
- ❖ Allows to train a robust feature extractor on various datasets.
- ❖ Relevant features are extracted to perform EEG signal classification in setups with very few labeled examples.
- ❖ Cross-Dataset FSL can significantly reduce the amount data to annotate for a new subject to achieve a target level of performance.

We think it might help designing BCI systems that are faster to calibrate to a new user

References

- [1] Xia Wei, A. Aldo Faisal, Moritz Grosse-Wentrup, Alexandre Gramfort, Sylvain Chevallier, Vinay Jayaram, Camille Jeunet, Stylianos Bakas, Siegfried Ludwig, Konstantinos Barmas, Mehdi Bahri, Yannis Panagakis, Nikolaos A. Laskaris, Dimitrios A. Adamos, Stefanos Zafeiriou, William Duong, Stephen M. Gordon, Vernon J. Lawhern, Maciej Sliwowski, Vincent Rouanne, and Piotr Tempczyk, "2021 beetl competition: Advancing transfer learning for subject independence & heterogenous eeg data sets," ArXiv, vol. abs/2202.12950, 2021.
- [2] He He and Dongrui Wu, "Transfer learning for brain-computer interfaces: A euclidean space data alignment approach," IEEE Transactions on Biomedical Engineering, vol. 67, no. 2, pp. 399–410, 2019.

Acknowledgements

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